PROJECT GUIDE · 2026-11-01

ecg-trust

A 12-lead ECG classifier that ships with evidence for *when it can be trusted*: patient-level evaluation, calibration, calibrated abstention, subgroup and noise robustness, external validation, and a comparison with a commercial algorithm.

PTB-XL · 21,799 ECGs | External · 28,646 ECGs | PyTorch · CPU | Conformal prediction

pending
macro-AUROC, internal test (95% CI pending)

pending
macro-AUROC, external validation (95% CI pending)

pending
macro ECE on test, before → after temperature scaling

pending
ECGs referred at α = 0.10 (internal test)

Author: Rittik Patra. Research software — **not a medical device** and not for diagnostic use. All numbers in this guide are read from the saved, version-controlled reports in `ecg-trust/reports/`; none are typed by hand.

Contents

1. Executive summary
2. Objective and motivation
3. Plan and decision log
4. Data
5. Methods
6. Results
7. Limitations and non-claims
8. Reproducibility
9. References

1 · Executive summary

Most ECG deep-learning portfolios stop at one number: the AUROC on a held-out split. A clinical team deciding whether to deploy a model needs more: are the probabilities honest, can the model say "I don't know", does it hold up on noisy recordings, on other hospitals and on patient subgroups, and is it better than the interpretation software already built into the ECG cart? This project answers those questions for a five-class 12-lead ECG classifier trained on PTB-XL.

> **Headline.** Final test and external results are pending the locked evaluation runs.

Question	How it is answered	Result
Does it rank cases well?	Patient-bootstrap AUROC, internal test	pending
Are probabilities honest?	ECE before → after temperature scaling	pending
Can it say 'I don't know'?	Label-conditional conformal refer rate, $\alpha = 0.10$	pending
Does it transfer?	External database, strict label map	pending
Is it robust to noise?	Synthetic noise at exact SNR; refer rate vs SNR	see §6.4
Better than the cart software?	Sensitivity at GE 12SL's specificity	see §6.7

2 · Objective and motivation

Objective. Build, evaluate and serve a classifier for the five PTB-XL diagnostic superclasses — normal ECG (NORM), myocardial infarction (MI), ST/T change (STTC), conduction disturbance (CD) and hypertrophy (HYP) — from 10-second, 12-lead ECGs, and produce evidence for when its output can be trusted.

Why trust is the product. A diagnostic-support model is only useful if a clinician can act on its output. Three properties matter as much as ranking performance. *Calibration*: a predicted 70 % should be right about 70 % of the time, otherwise thresholds and risk communication break. *Abstention*: on ambiguous or poor-quality recordings the model should refer to a cardiologist instead of guessing. *Transportability*: performance on a different population, device and labelling process is usually lower; the size of the drop must be measured, not assumed.

Scope. Multi-label classification at 100 Hz. The 500 Hz recordings, the 71 individual statements, new architectures and clinical thresholds are out of scope. Compute is a 4-core CPU container, so models are deliberately compact.

What a reader should take from this project

Data rigour
Checksum-verified downloads, audited labels, patient-level splits with an automated leakage test.

Honest evaluation
One-shot, locked test set; patient-bootstrap CIs; an external dataset from another country.

Trust layer
Temperature scaling and per-class conformal "present / absent / refer" decisions.

Engineering
Tested package, CLI, CI, FastAPI service, Streamlit demo, Docker, model card.

3 · Plan and decision log

The project followed a written design specification and a 24-task, test-driven implementation plan with four milestones. Each milestone ended at a review gate with the author.

Milestone	Deliverable	Gate
M1 · Data	Downloader, labels, splits, signal caches, external loader, feature baseline, data audit	Audit and label-map review
M2 · Models	1-D ResNet, 3-seed ensemble, calibration, conformal, robustness, 12SL comparator — validation data only	Freeze: models, thresholds and maps hashed
M3 · Final evaluation	Single logged run on the internal test fold and on the external set	Results review
M4 · Product	API, demo, Docker, CI, model card, README, this guide	Portfolio-ready

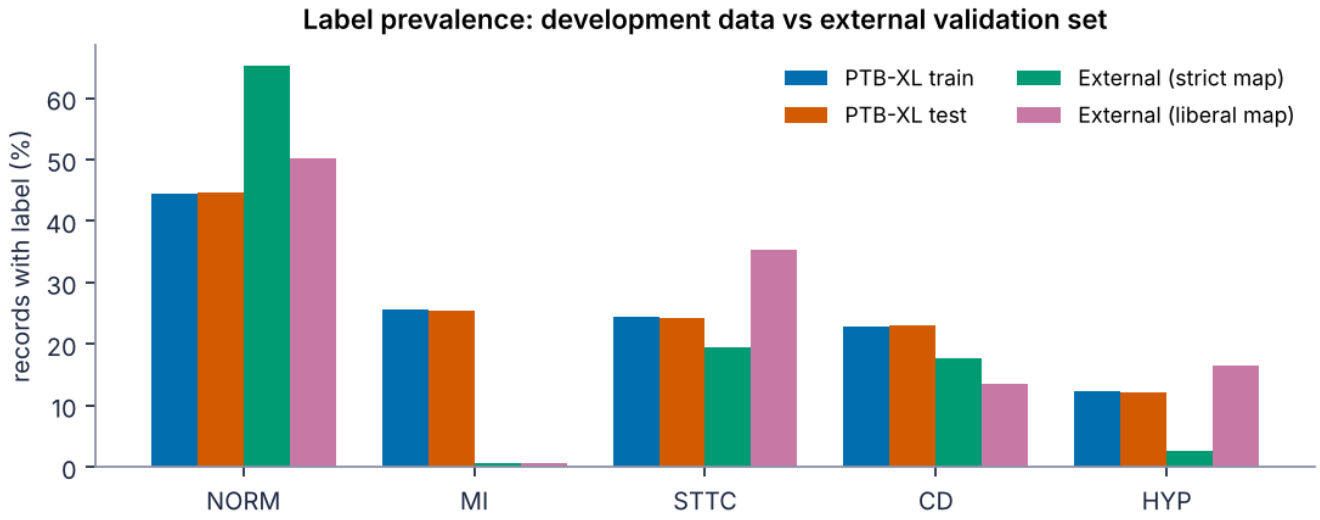
Decisions taken during the project

Decision	Why
PhysioNet AWS open-data mirror, every file SHA-256-verified, resumable downloads	physionet.org was not reachable; the proxy cut long transfers
Patient-level split of fold 9 into model-selection and calibration halves	conformal guarantees need calibration data unused for training or tuning
External label map completed to all 94 SNOMED codes; strict PTB-XL-faithful primary map, liberal map as sensitivity analysis	the dataset's condition list missed 43 codes, incl. complete RBBB and LBBB; PTB-XL treats isolated ST/T findings as form statements
NORM proxy = sinus-family rhythms only; unlabelled external records excluded	mirrors PTB-XL, where sinus bradycardia can be a normal ECG
1-D ResNet width 64 (1.49 M parameters), 3 seeds	author's choice after a CPU timing benchmark
Synthetic, SNR-exact noise instead of NSTDB records	NSTDB was not on the reachable mirror
GE 12SL map built from statement names cross-checked with PTB-XL+ SNOMED concepts; ambiguous statements unmapped	keeps the commercial comparison conservative
Freeze manifest + single-use test log	prevents tuning on the test sets

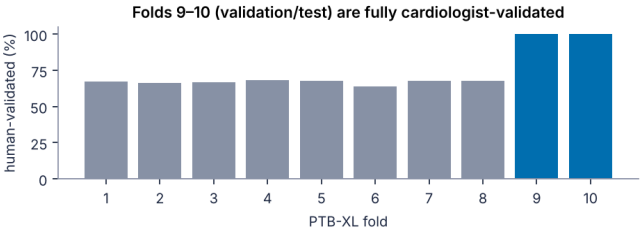
4 · Data

Dataset	Role	Size
PTB-XL 1.0.3 (Wagner et al., 2020)	Training, validation, internal test	21,799 ECGs, 18,869 patients, 10 s, 12 leads
PTB-XL+ 1.0.1 (Strodthoff et al., 2023)	Feature baseline; GE 12SL machine statements	ECG measurements from GE 12SL, Glasgow Uni-G and ECGDeli
Large-scale 12-lead arrhythmia database 1.0.0 (Zheng et al., 2022)	External validation	45,152 ECGs from Chapman University / Shaoxing / Ningbo hospitals; 28,646 evaluated

All files were downloaded from the PhysioNet open-data mirror and verified against the published SHA-256 checksums (134,363 files, 6.2 GB, zero failures). Labels follow the published benchmark convention: a superclass is positive if any diagnostic SCP statement maps to it. 411 PTB-XL records have no diagnostic superclass and are excluded.



Label prevalence. The external set is dominated by sinus-rhythm-only ECGs (the NORM proxy) and has very few MI labels — a large, deliberate dataset shift.



Share of labels validated by a cardiologist per PTB-XL fold. Evaluation folds are cleaner than training folds.

Splits (patient-level)

train	17,084 records
val_select	1,072 records
val_calib	1,074 records
test	2,158 records

Folds 1–8 train; fold 9 split by patient into model selection and calibration halves; fold 10 is the locked test set. No patient appears in two splits (checked by a unit test and at run time).

Harmonising the external labels

The external database codes diagnoses in SNOMED CT; 43 of the 94 codes present in its records were missing from its own condition list and were identified from the PhysioNet/CinC Challenge 2021 mapping. The primary map is strict and PTB-XL-faithful: isolated ST/T-wave findings and voltage-only LVH criteria are *form* statements in PTB-XL and are therefore not mapped. A liberal map that includes them is reported as a sensitivity analysis. NORM is a proxy (sinus-family rhythm codes only), and records without any superclass are excluded, exactly as in PTB-XL.

5 · Methods

1 · Signals
100 Hz, 12 × 1000, mV; per-lead standardisation fitted on train only.

2 · Models
Gradient boosting on PTB-XL+ features; compact 1-D ResNet × 3 seeds.

3 · Calibration
Per-class temperature fitted on the calibration half of fold 9.

4 · Abstention
Per-class Mondrian conformal sets → present / absent / refer.

5 · Evaluation
Locked one-shot test and external runs, bootstrap CIs, subgroups, noise.

Models

Feature baseline. One histogram gradient-boosting classifier per class on 757 GE 12SL measurements (intervals, amplitudes, axes), with ECGDeli features as an open-source alternative. **Deep model.** A 1-D ResNet (width 64, stages [2, 2, 2, 2], kernel 7) trained with AdamW, one-cycle learning rate and early stopping on validation macro-AUROC; three seeds are averaged as a deep ensemble (Lakshminarayanan et al., 2017).

Calibration

For each class a single temperature T rescales every member's logit before averaging: $p = \text{mean}_m\ \sigma(z_m / T)$, with T chosen to minimise the negative log-likelihood on the calibration split (Guo et al., 2017). Calibration is measured with the expected calibration error (15 equal-mass bins), Brier score and reliability diagrams.

Conformal abstention

For each class, non-conformity scores are 1 − p for positive and p for negative calibration ECGs, and each label value gets its own finite-sample quantile (label-conditional, "Mondrian" split conformal; Vovk et al., 2005; Angelopoulos & Bates, 2021):

$$\hat{q}_y = \text{the } \lceil (n_y + 1)(1 - \alpha) \rceil\text{-th smallest score among calibration ECGs with label } y$$

A new ECG's set contains "present" if $1 - p \le \hat{q}_1$ and "absent" if $p \le \hat{q}_0$. A single-element set is a decision; otherwise the ECG is referred. Under exchangeability the true label is covered with probability ≥ $1 - \alpha$ separately for positives and negatives.

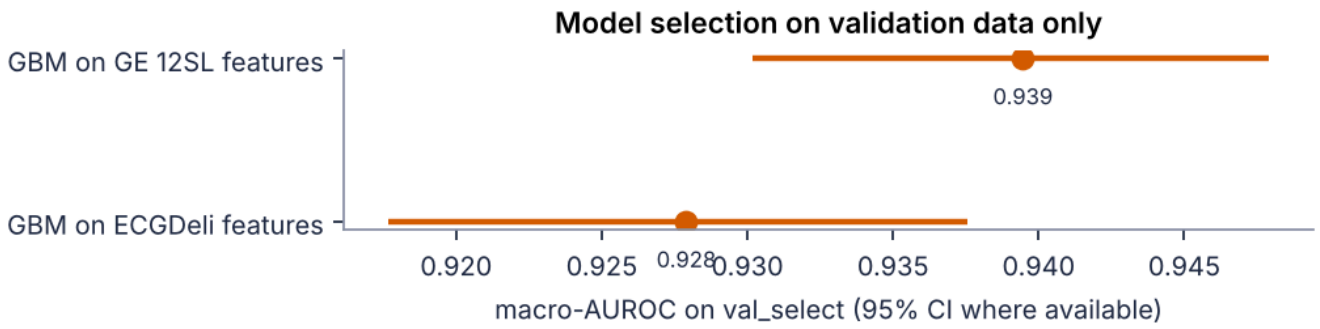
> **Caveat.** The guarantee is marginal and assumes new data look like the calibration data. The external set violates this on purpose; its coverage is reported to show by how much.

Robustness, subgroups and comparator

Synthetic noise mimicking the MIT-BIH Noise Stress Test types (baseline wander below 0.5 Hz, EMG-like 20–45 Hz, electrode-motion steps and bursts) is added at exact per-lead SNRs from 24 to −6 dB, plus lead dropout. Subgroups (sex, age band, device, PTB-XL noise flags, pacemaker) are descriptive, with patient-bootstrap CIs; groups with fewer than 30 positives or negatives are marked insufficient. The GE 12SL comparison uses 12SL's own operating point: the model's sensitivity is read at 12SL's specificity on the same ECGs, using a published statement-to-superclass map.

6 · Results

6.1 Model selection (validation data only)



All modelling choices were made on val_select before the freeze. Points are macro-AUROC; bars are patient-bootstrap 95% CIs.

6.2 Internal test (PTB-XL fold 10, single locked run)

Internal test results appear here after the single locked test run.

Figure pending: ROC curves on the internal test fold.

Figure pending: Reliability after temperature scaling.

Figure pending: Reliability before scaling, for comparison.

6.3 Calibrated abstention

Conformal coverage and refer rates appear here after the test run.

6.4 Robustness to noise

Figure pending: Macro-AUROC and conformal refer rate as synthetic noise increases (left to right). A trustworthy system should refer more ECGs as quality falls.

6.5 Which leads matter

Figure pending: AUROC drop when a single lead is zeroed. Descriptive only — not a mechanism.

6.6 Subgroups

Figure pending: Descriptive subgroup AUROCs with patient-bootstrap 95% CIs (internal test). Differences are only noted when intervals do not overlap.

6.7 Comparison with GE 12SL

Figure pending: Model sensitivity at 12SL's own specificity on the same test ECGs. Indicative: depends on the published statement map.

6.8 External validation

External results appear here after the single locked external run.

Figure pending: Per-class AUROC and calibration error, internal test vs external validation.

7 · Limitations and non-claims

- **Not a medical device.** The model is research software, not clinically validated, and must not be used for diagnosis.
- **Noisy reference standard.** PTB-XL labels come from reports; about a third of training labels were not validated by a cardiologist.
- **Marginal guarantees.** Conformal coverage holds on average under exchangeability, not for every patient, and not under dataset shift.
- **Descriptive subgroups.** Many subgroups are compared; small groups are flagged and differences are not causal.
- **Label harmonisation.** External labels are mapped, and the NORM proxy is imperfect; the sensitivity map shows how much this matters.
- **Synthetic noise.** The real NSTDB noise records were not reachable from the compute environment; synthetic noise is parameterised to mimic them.
- **Comparator.** The 12SL comparison depends on a statement map and on 12SL's binary output; it is indicative.

8 · Reproducibility

```
pip install torch --index-url https://download.pytorch.org/whl/cpu
pip install -e ".[serve,dev]"
ecgtrust download --dataset ptbxl && ecgtrust download --dataset ptbxl_plus && ecgtrust download
--dataset external
ecgtrust audit
ecgtrust train --model baseline --features 12sl
ecgtrust train --model resnet --seeds 0 1 2
ecgtrust evaluate --split val_select
ecgtrust freeze                          # hashes checkpoints, config and label maps
ecgtrust evaluate --split test           # single use, logged
ecgtrust evaluate --split external       # single use, logged
ecgtrust guide                           # rebuilds this PDF from reports/
```

Freeze manifest: `not frozen y`. Seeds 0, 1, 2. Python 3.13.18, PyTorch 2.15.0. Every unit test runs in CI on synthetic fixtures without network access.

9 · References

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